The Trade-off Was in the Labels: Causal Supervision for Turn-Aware Streaming ASR

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Abstract

Every voice agent must decide, in real time, whether the user has finished talking. The deployed default decides from acoustics alone, using a voice-activity detector plus a silence timeout, while frontier commercial recognizers have moved the decision into the ASR model, where the words are; every member of this *turn-aware* class is closed. We present, to our knowledge, the first open turn-aware streaming ASR system with a published training recipe: a small LoRA adapter on Qwen3-ASR-0.6B, trained in hours on one GPU, that transcribes, detects end-of-turn semantically, handles dictation, and grounds transcription in session context. On a deployment-matched streaming benchmark it reaches 0.97 boundary recall at 0.39 s median latency with 0.3 false fires per speech-minute, replicated on a fresh test set; no silence timeout reaches this operating point. The recipe rests on one principle: every label for a streaming decision must be computable from the input up to the decision point. Supervision inherited from offline corpora violates it, and the resulting *clairvoyant* labels manufactured, across eight controlled supervision compositions, training oscillation and a phantom recall-versus-precision frontier: appending one second of silence to the evaluation clips raised a "broken" model's fire recall from 0.10 to 1.00. The same leak recurred on the context axis: a biasing prefix became a copyable shortcut (40% wrong-profile intrusion); counterfactual examples where context and audio disagree cut intrusion to 0.8% while keeping most of a +28.9 pp entity-recall benefit. Weights, recipe, and benchmark are released.

Code: `github.com/19PINE-AI/turn-aware-asr` • Website: `01.me/research/turn-aware-asr`

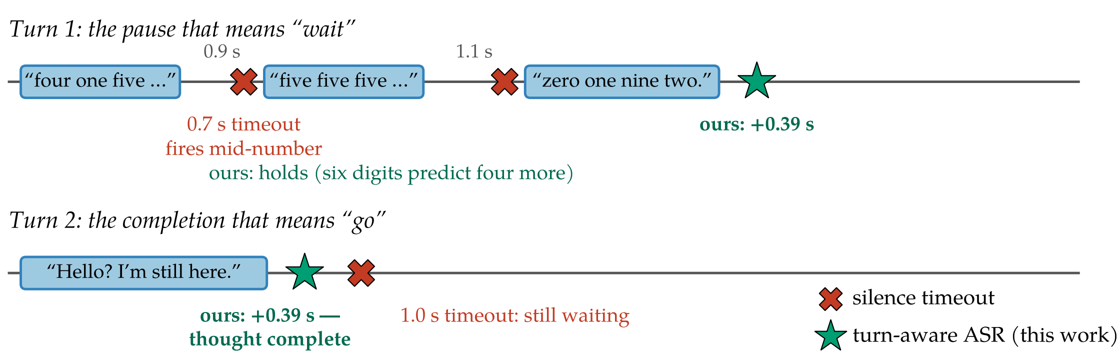

Figure 1. No timeout gets both turns right. At 0.7 s it interrupts the dictated number twice; at 1.0 s it still keeps the user waiting after "I'm still here." Ours holds through the pauses and fires 0.39 s after the thought completes.

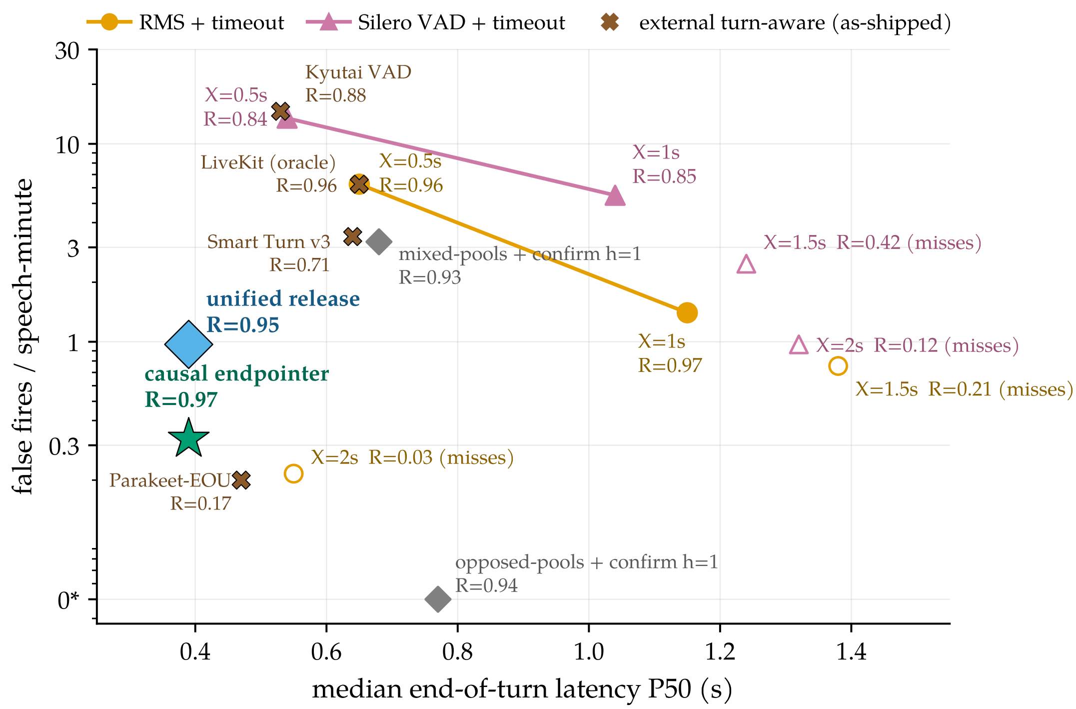

Figure 2. The causal model escapes the timeout curve. Median end-of-turn latency vs. false fires (log scale; 0* plotted at 0.05) on identical audio, detector, and scoring. A silence timeout can only slide along its curve; the causally supervised model sits strictly inside the whole family because it reads completeness, firing 0.39 s after a finished thought while holding through a 1.4 s mid-sentence pause.

1 Introduction

A voice agent that cannot tell when you have finished speaking fails in two symmetric ways: it interrupts you mid-thought, or it stares at you after you stop. The industry default delegates the decision to a voice-activity detector (VAD) plus a fixed silence timeout, and this cascade fails structurally. Humans exchange turns with median gaps around 200 ms [Stivers et al., 2009, Levinson and Torreira, 2015], deployed timeouts sit at 500–1000 ms [Skantze, 2021], and pauses *inside* turns routinely exceed pauses between turns, so no threshold is both fast and safe. The information that resolves the ambiguity (is this thought complete?) is in the words, and the words live inside the recognizer.

Frontier commercial systems have drawn that conclusion and moved end-of-turn into the ASR model itself: Deepgram's Flux emits end-of-turn events from the recognizer's forward pass [Deepgram, 2026], and OpenAI's Realtime API waits longer after a trailing "ummm" than after a completed sentence [OpenAI, 2025]. We call this class **turn-aware streaming ASR**: one model that transcribes while deciding, from semantics and silence jointly, whether the turn is over. Every commercial member of the class is closed, and the open landscape offers only pieces: weights with no training account (Kyutai STT, Kyutai Labs, 2025; Parakeet-EOU, NVIDIA, 2026) and turn *detectors* that do not transcribe (Smart Turn, Pipecat team, 2025; LiveKit, LiveKit, 2025; TEN, TEN Team, 2025).

This paper presents, to our knowledge, the first open member of the class, with the complete training recipe. The system is deliberately small: a LoRA adapter [Hu et al., 2022] on Qwen3-ASR-0.6B [Shi et al., 2026], trained in hours on one GPU from roughly twenty thousand synthesized examples. One checkpoint transcribes incrementally, emits an in-transcript end-of-turn token, holds through dictated digit strings, and grounds transcription in a session context prefix. On a deployment-matched streaming benchmark, and again on a fresh test set drawn after development ended, it reaches an operating point that no silence timeout and none of the open turn-aware systems attain: sub-half-second median latency with higher boundary recall and fewer false fires (Figure 2). The same prefix substantially lifts entity recall on entity-dense audio without degrading transcription quality. Weights, recipe, and benchmark are released.

The recipe's substance is not architectural: the entire turn capability comes from the labels, and the paper's central finding is what those labels must satisfy. Supervision inherited from offline corpora defines end-of-turn using audio *after* the decision point, audio a streaming model can never see: offline clips are cut by forced alignment at exactly the boundaries a streaming model must detect, and offline transcripts record who spoke next. We call such labels *clairvoyant*.

Clairvoyant labels manufacture phantom trade-offs. In our development record, eight successive supervision compositions with identical architecture, adapter, and data volume traced what looked like a fundamental recall-versus-precision frontier, oscillating between checkpoints that fired eagerly mid-utterance and checkpoints that barely fired at all. A one-variable intervention exposed the frontier as an artifact: appending one second of silence to the evaluation clips took a "broken" checkpoint's fire recall from 0.10 to 1.00. One causal labeling rule (fire iff the words so far are semantically complete and at least 0.3 s of silence has been observed) plus a minimal-pair construction makes training monotone, with nothing else changed, and yields a single checkpoint that dominates all eight predecessors.

The principle is not specific to time. Trained only on examples where the biasing prefix matched the audio, the model learned to *copy* the prefix rather than listen: a wrong-user profile wrote the wrong entity into the transcript 40% of the time. The leak has the same shape (a training target that depends on a signal the model cannot trust at decision time) and the same repair: counterfactual examples in which context and audio deliberately disagree and the target follows the audio, which drive intrusion below 1% while largely preserving the biasing benefit. Two instances of one failure class, cured by the same construction, ground a checklist for any streaming decision trained from offline logs.

Our contributions:

1. **The first open turn-aware streaming ASR system, with its training recipe and benchmark.** One labeling rule and two pair constructions yield a single small model that transcribes, emits in-transcript end-of-turn events, holds through dictation, and grounds transcription in context, beating every silence-timeout setting. Weights, recipe, and a deployment-matched benchmark, under which the historical model ranking *inverts*, are released.

2. **A failure class, with a diagnosis by intervention.** Clairvoyant labels (streaming supervision that depends on post-decision input) manufacture training oscillation and phantom capability trade-offs. A one-variable experiment falsifies the frontier that eight supervision compositions had appeared to trace; a synthetic study isolates the mechanism.

3. **Evidence that the principle generalizes.** The contextual analog of the temporal leak, context copying under matched-only training, is cured by the analogous counterfactual

construction, folding dictation and context grounding into the same checkpoint. Both instances distill into a checklist for streaming supervision pipelines.

2 Problem formulation and system overview

Two turns bound the problem from opposite sides (Figure 1). In the first, the silences carry no evidence that the turn is over but the words do (six digits of a ten-digit pattern predict continuation), yet a 0.7 s timeout fires twice mid-number, a failure production teams report exactly [LiveKit, 2026]. In the second, the thought is complete the instant the last word ends and a human would respond within roughly 200 ms [Stivers et al., 2009], yet a 1.0 s timeout makes the user wait five times that long. The first case requires holding *through* long silence, the second firing at near-zero silence, and no single threshold can do both, because the gap between the cases is not measurable in seconds of quiet. On our conversational benchmark the silence-gap distributions of within-turn pauses and between-turn boundaries overlap almost completely, with the pauses the *longer* of the two (§6); the same overlap and the resulting dilemma have defined cascade endpointing since Raux and Eskenazi [2008] [Skantze, 2021]. The resolving signal is semantic completeness, which is exactly what a recognizer computes on the way to a transcript.

Transcription has its own version of the argument. A voice agent hears names, tickers, and account emails: tokens nearly unrecoverable from acoustics alone but obvious given who is calling about what. Every major commercial STT ships a context mechanism, yet none publishes controlled uplift numbers, and the open model we build on describes its context facility in one qualitative sentence [Shi et al., 2026]. A turn-aware model with a context slot can ground the whole session's transcription in the profile the agent's user memory already maintains [Li, 2026]; §5 measures exactly this.

The system under study. A single-channel audio stream arrives in 0.5 s chunks. At each chunk boundary the system must extend a running transcript and decide whether the turn has ended, an irreversible and latency-sensitive event. A text context prefix may be supplied at session start. We start from Qwen3-ASR-0.6B [Shi et al., 2026], an open unified speech model (AuT audio encoder + dense Qwen3 decoder) with strong offline WER but no endpointing, claim two reserved vocabulary slots as marker tokens <EAGER_END_SPEECH> and <END_SPEECH>, and LoRA-fine-tune [Hu et al., 2022] the decoder to emit them inside the transcript at endpoint moments (rank 16 for the endpointing experiments, 32 for the released unified model; adapter details in Appendix G). Training data is synthesized from LibriSpeech [Panayotov et al., 2015] and AMI [Carletta et al., 2005]; training takes hours on one GPU. At inference, three interpretable knobs sit outside the model (Figure 3): an RMS **energy gate** that never *starts* a segment on a silent chunk, a **confirm horizon** h that holds a fire candidate for h further silent chunks before signaling END, and a **max-segment force-flush** that bounds the committed transcript segment. Every reported arm states its knob settings. Table 6 (Appendix C) places the system in the landscape of open and commercial turn-aware systems.

3 Causal supervision

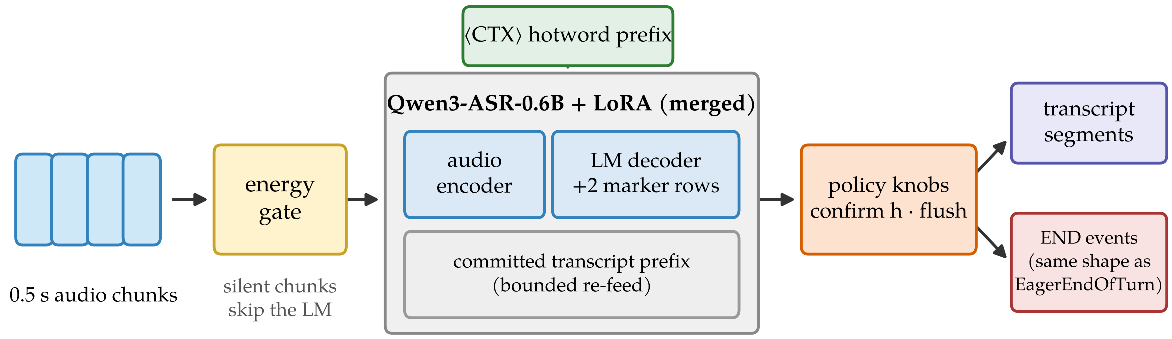

Figure 3. System shape. Audio chunks pass an energy gate; the merged LoRA model transcribes the current bounded segment with the `<CTX>` prefix in the system slot and may emit marker tokens; two policy knobs sit behind the model. Outputs are transcript segments plus END events.

3.1 Clairvoyant labels and the causality principle

A streaming model may act at time t only on the input prefix up to t. The constraint sounds too obvious to state, yet supervision built from offline corpora violates it systematically, because offline data encodes the future in two places. First, offline ASR corpora are segmented by forced alignment, so clips are cut at the end of speech; endpoint labels or evaluations built on such clips demand a fire on audio that ends *during or immediately after* speech, a condition deployment never presents, because on a live channel silence keeps arriving after a real speaker stops. Second, offline transcripts record who spoke next, so pause labels derived from segmentation (same speaker continues: disfluency, hold; another speaker follows: boundary, fire) are functions of the future; measured at the decision point, the two classes are statistically indistinguishable (§6). We give the failure class a name:

Definition: clairvoyant labels

A label for a streaming decision at time t is *clairvoyant* if its value depends on input after t. Pools containing clairvoyant labels do not merely add noise: because the dependence is systematic, they partition the data into internally consistent subsets with incompatible optima. The observable symptoms are (a) training-time oscillation between mode attractors and (b) apparent capability trade-offs across checkpoints, a phantom Pareto frontier. **Principle: every training label and every evaluation target for a streaming decision must be computable from the input up to the decision point.**

Clairvoyance is target leakage transposed into time, and the class is wider than time: the same structure appears whenever the target correlates with any signal the model cannot *trust* at decision time. A context prefix that always matches the audio during training is such a signal, because at deployment the profile can simply be wrong (§3.3); Figure 7 (Appendix A) shows the two instances side by side. In both coordinates the cure is a construction rather than a loss term: minimal pairs or counterfactual twins that make the untrustworthy signal uninformative by design, so the only policy consistent with the training data is the one that keys on observables. §6 presents the evidence. Figure 4(a) grounds the definition in real benchmark audio.

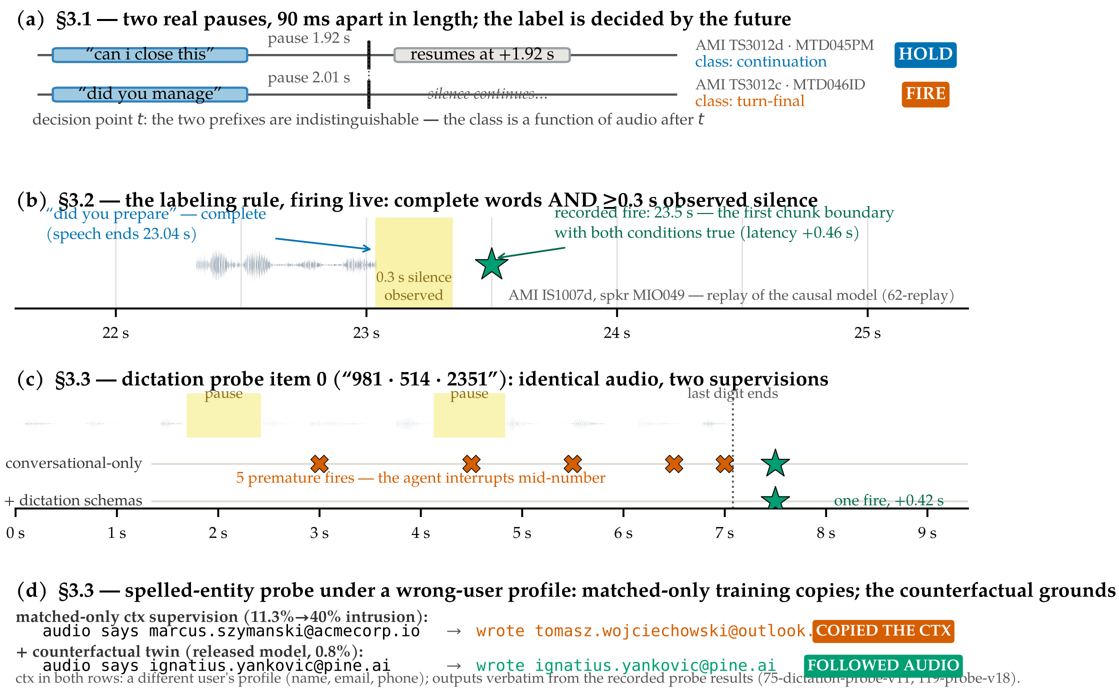

Figure 4. One real example per subsection, all verbatim from the benchmark material and recorded result logs. (a) §3.1: two held-out AMI pauses 90 ms apart; at the decision point the prefixes are indistinguishable, so the class is a function of audio after t, exactly what a clairvoyant label encodes. (b) §3.2: "did you prepare" is complete at 23.04 s, 0.3 s of silence is observed by 23.34 s, and the recorded fire lands at the next chunk boundary (+0.46 s). (c) §3.3, dictation: identical audio under two supervisions; conversational-only fires five times inside the number, the dictation schemas fire once, +0.42 s after the last digit. (d) §3.3, context: under a wrong-user profile the matched-only model writes the profile's email verbatim; the counterfactually trained release follows the audio.

3.2 Supervising end-of-turn detection

The entire turn capability comes from the labels. One rule generates every training example: *emit <EAGER_END_SPEECH><END_SPEECH> at a point iff the speech so far is semantically complete AND $\geq$0.3 s of silence has elapsed since the last speech.* Both conditions are functions of the audio prefix. Completeness is decided by a transcript heuristic (trailing filler or connective, mid-word cut, or very short non-acknowledgment ⇒ incomplete; Appendix A); silence is decided by the synthesized waveform itself.

Two constructions carry the semantics-versus-silence discrimination that §2 asked for (Table 1). The **minimal pair** (schemas 1–2; Figure 8, Appendix A) presents the same complete utterance twice, with and without an observable silence tail, with opposite targets: completeness alone must not fire; the model has to *see* the silence. This is the direct antidote to supervision under which silence becomes optional (row 4 of the composition record, Table 4). The **pause pair** (schemas 4–5) presents two-utterance sequences whose gap is bridged or fired depending only on whether the words before the pause are complete: "…and then, *(pause)*" holds; "…that's everything. *(pause)*" fires, even if the speaker later resumes. Quick resumes after a legitimate fire become two transcript segments, a measured cost rather than an error (§4.1), because at decision time the correct answer does not exist on the channel. Schemas 7–8 cover the silence-heavy channel shape that pure-utterance corpora never show, and pair

Table 1. The eight training schemas (∼12k examples; LibriSpeech and AMI roughly 50/50). M = <EAGER_END_SPEECH><END_SPEECH>.

#	Construction	Target
1	complete utterance + 0.3–1.2 s silence tail	`text` M
2	*same utterances as #1*, tail ≤0.1 s	`text`
3	speech truncated mid-utterance at 30–80%	partial `text`
4	complete A + gap 0.3–2.5 s + B (+ tail)	`A` M `B` M / `A` M `B`
5	*incomplete* A + gap + B + tail	`A B` M
6	complete utterance + long silence (2–4 s)	`text` M
7	pure silence 0.5–4 s	(empty)
8	leading silence 0.5–2 s + utterance + tail	`text` M

examples mix same- and different-speaker sources, so the fire decision cannot key on speaker identity, which single-channel deployment never observes. An iso-count ablation isolates what each construction buys and confirms each predicted failure (Appendix D, Table 11). Figure 4(b) shows the rule firing in deployment-matched replay.

3.3 Supervising dictation and context biasing

The same discipline extends the recipe to the two transcription behaviors that §2 motivated. **Dictation:** partial phone numbers followed by a pause are labeled *hold*, because the pattern predicts continuation; complete numbers plus observed silence fire. **Spelled entities and context:** synthesized utterances spelling uncommon names and email addresses are targeted in normalized written form, half with a user profile in the context slot. These schemas extend the endpointing pool to ∼15k examples for retraining the same LoRA.

Matched-context supervision alone, however, reintroduces the §3.1 failure class on the context axis. If the profile always agrees with the audio, the target is predictable from the prefix without listening, and the model learns that the context *is* the answer: it copies whatever entity the prefix names (§5 measures the consequence). The **counterfactual twin** removes the shortcut: a third of the spelled examples carry a *conflicting* profile, where the context names one identity, the audio spells another, and the target follows the audio, exactly as the minimal pair presents the same utterance with and without a silence tail. A second counterfactual of the same shape covers natural-speech biasing: the context is an entity list that disagrees with the audio, and the target again follows the audio. The released pool (∼20k examples) folds in both counterfactuals plus a 20% plain-transcription replay fraction that anchors offline WER (Appendix G); the released adapter is rank 32, which §5 shows is what the added breadth costs. Figure 4(c,d) shows both fixes on recorded probe outputs.

4 Evaluation protocol

4.1 Streaming replay benchmark

Endpointing is evaluated by streaming replay, and the causality principle applies to the benchmark itself: an evaluation whose clips end at forced-alignment cuts demands behavior on a condition deployment never produces, and §6 shows such an evaluation actively misdirects development.

Material and ground truth. Continuous single-channel stretches from held-out AMI meetings: one target speaker's utterances at their true timeline offsets, silence between them, 30–60 s per stretch. This is the voice-assistant shape, where audio never ends at a speech boundary. Each utterance-end boundary is classified by what is observable *on this channel*: **turn-final** (same-speaker gap ≥ 2 s), where the system should fire; **continuation** (gap in $[0.3, 2)$ s, same speaker resumes), where firing is a measured segmentation cost rather than an error, because the correct answer depends on the future; and **internal** (pauses inside utterances, or <0.3 s after speech), where firing is a false fire. Metrics: boundary recall within $[-0.25, +1.5]$ s, false fires per speech-minute, latency percentiles, resume rate, streaming WER of flushed transcripts, and per-chunk compute. The development set has 25 stretches; a twice-larger confirmation set was drawn after all development ended, and a 100-stretch held-out set is used for the released model. An early version of the benchmark itself contained a clairvoyant rule, caught by our own checklist; all numbers use the corrected classification (Appendix B).

Why comparisons run behind the energy gate. Models trained only on speech-initial examples hallucinate markers on silence at roughly two fires per second, so *ungated* recall is uninterpretable: a random 1.9 Hz spammer scores 0.96 against the tolerance window (Appendix B, Figure 9). All comparisons therefore run behind the energy gate; the causal recipe additionally bakes silence schemas into training.

4.2 Probes and offline evaluations

The two §2 scenarios get probes, both run through the exact streaming stack of §4.1. The **dictation probe** replays ten-digit phone numbers (3-3-4 groups from held-out Free Spoken Digit Dataset speakers [Jackson et al., 2018], with inter-group pauses) and scores premature fires per number, final-boundary recall, and digit accuracy. The **spelled-entity probe** decodes synthesized utterances spelling uncommon names and email addresses, with and without a user-profile prefix, plus a wrong-user *distractor* profile, and scores exact match and wrong-profile intrusion. Both probes were enlarged after development ended; released-model numbers use the enlarged versions. **Context biasing** on natural speech is measured on Earnings-22 [Del Rio et al., 2022], spontaneous earnings-call speech dense in tickers, executives, and products, with entities LLM-extracted and filtered to genuine proper nouns, as entity recall with and without a relevant hotword prefix. **Offline WER** is scored on the full official LibriSpeech splits, and **serving** is measured on stock vLLM rather than a simulator (Appendix H).

5 Results

5.1 End-of-turn detection

Table 2 reports the main result. Against the strongest prior supervision compositions (each at its best knob setting), the causal model with the gate alone is better on every axis at once: its *raw* false-fire rate beats the mixed-pools model's *confirmed* rate, at half the latency and higher recall. Against the deployed cascade, the comparison is structural. A timeout must be long enough to survive within-turn pauses, so it can only trade latency against false fires along one curve; the causal model escapes the curve by reading completeness, combining exactly the

Table 2. Main endpointing result (deployment-matched replay, energy gate on; the latency-vs-false-fire view is Figure 2). Rows 1–4: the 25-stretch development benchmark. Last row: the fresh twice-larger confirmation set drawn after all development ended. WER is a fraction; means can exceed 1 through insertions. †WER mean dominated by long-monologue repetition loops when no force-flush bounds the segment (WER *median* 0.30 on both stretch sets; the force-flush of Appendix H bounds this tail).

Policy	Recall ↑	P50 ↓	P95 ↓	False/min ↓	WER_{mean} ↓
mixed-pools + gate + confirm h=1	0.927	0.68 s	1.27 s	3.2	0.36
opposed-pools + gate + confirm h=1	0.938	0.77 s	1.06 s	0.0	4.96†
causal + gate (ours)	**0.969**	**0.39 s**	**0.70 s**	**0.3**	1.43†
causal + gate + confirm h=1	0.969	0.89 s	1.20 s	0.0	1.43†
causal + gate, fresh 50-stretch set	0.924	0.42 s	0.70 s	0.2	4.63†

pair of behaviors §2 argued no threshold can combine. Sweeping the full timeout family over two detectors and all settings on both stretch sets, no setting reaches its latency-and-false-fire corner at any recall (Appendix D). The inference crutches also retire: the confirm horizon that earlier compositions needed cancels only 3 fire candidates for the causal model versus 241 for the mixed-pools model, so the phrase-versus-turn discrimination moved from inference policy into the weights. The result survives fresh data (Table 2, last row): latency and false fires hold, and the small recall dip is within sampling variation (Appendix D). By design, quick resumes after a legitimate fire are always segmented; deployments preferring merged segments pay +0.5 s via the confirm horizon.

External turn-aware systems on the same protocol. We ran the open turn-aware systems of §8 through the identical benchmark as-shipped, sweeping their public thresholds rather than retraining (Figure 2). None reaches the causal model's corner: the standalone detectors are gated on VAD silence by construction and slide along the timeout curve, LiveKit's text detector reaches high recall only with an oracle transcript and many times our false-fire rate, and the open recognizers with turn signals either rarely fire (Parakeet-EOU) or fire spuriously (Kyutai). To our knowledge this is the first measurement of the open turn-aware class on a common protocol; it is as-shipped, not best-achievable, since none was tuned for AMI close-talk.

5.2 Context biasing and dictation

Biasing works, and survives session length. On Earnings-22, a relevant hotword prefix lifts the base model's entity recall from 66.7% to 95.6% while nudging WER slightly down: the prefix acts as a domain hint, not a distortion (Figure 5a). With up to 12 s of unrelated speech interposed between prefix and target words, the advantage stays flat (Figure 5b), which is the property a voice agent actually needs, since the user's account entities must help at minute five, not just in the first utterance. The base model's report describes this capability qualitatively without numbers [Shi et al., 2026]; to our knowledge these are the first on this model family [cf. Durmus et al., 2026].

Dictation, and the copying result. The conversational-only endpointing model of §5.1 fails both §4.2 probes exactly as §2 predicts: it reads a digit group as a finished thought, firing prematurely five times per dictated number, and spells emails at 4% exact (Table 3). The dictation schemas of §3.3 fix the former immediately. The context schemas, trained with

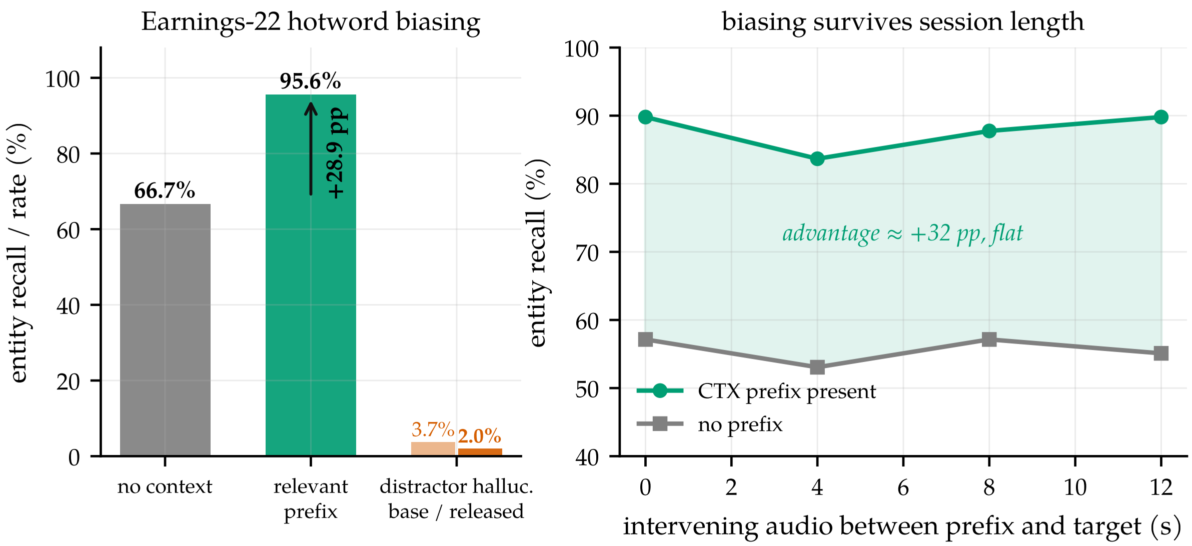

Figure 5. Context biasing, measured. (a) Earnings-22 base-model entity recall without context and with the relevant hotword prefix (+28.9 pp), and the distractor-prefix hallucination rate for the base model versus the released rank-32 model, whose natural-speech counterfactual drives it from 3.7% to 2.0%. (b) Recall vs. seconds of unrelated speech between prefix and target: the advantage does not decay with session age.

Table 3. Dictation and spelled-entity probes. Premature fires per dictated 10-digit number, final-boundary recall, digit accuracy; exact-match on spelled names/emails without/with the profile; wrong-profile intrusion. Conversational-only and released rows are scored on enlarged probes; the intermediate row uses the smaller original probes. Confidence intervals and a scoring-window sensitivity analysis are in Appendix D. The conversational-only model's nonzero intrusion is the base model's native context-following, which that fine-tune leaves untouched.

	Premat./seq ↓	Final rec. ↑	Digit acc ↑	Name −/+	Email −/+	Intrus. ↓
conversational only	4.96	0.82	0.889	0.25 / 0.82	0.04 / 0.58	1.7%
timeout X=0.5 s	2.00	1.00	–	–	–	–
timeout X=1.0 s	0.74	1.00	–	–	–	–
+ dictation (match-ctx)	0.36	0.78	0.996	1.00 / 1.00	0.03 / 0.93	11.3%
released (r32, unified)	**0.16**	**0.88**	**0.998**	**0.98 / 1.00**	0.11 / **0.93**	**0.8%**

matched profiles only, produce the predicted copying: the model writes whatever entity the prefix names, so a wrong-user profile puts the wrong email into the transcript, with intrusion rising from 11.3% to 40% as training lengthens. Adding the counterfactual twin drives intrusion to zero at every checkpoint of the run on the training probe, and to 0.8% (95% CI [0.2, 3.0]%) on the enlarged post-development probe, while leaving dictation and matching-context accuracy intact (Table 3): context becomes a prior, not a substitute for listening.

5.3 The unified model and its costs

The release is the merged unified checkpoint, the label-spec builder and training code, the probes, and the replay benchmark. The released checkpoint, a rank-32 adapter trained on one pool of ∼20k examples, endpoints conversation, holds through dictation better than every timeout setting, grounds spelled entities in context, and ignores a wrong profile (Table 3, last row), with no deployment-time checkpoint swap; on the 100-stretch held-out replay set it scores 0.982 boundary recall at 1.30 false fires per speech-minute, still strictly inside the timeout family of Figure 2. Its endpointing numbers across stretch sets, the confirm-horizon dial, a scoring-window sensitivity analysis, and three-seed retraining stability are collected in Appendix D.

Which frontiers are real. Folding four behaviors into one adapter costs endpointing precision: 0.97 false fires per speech-minute on the development set versus 0.3 for the pure-endpointing checkpoint, a genuine capacity trade-off we report and do not tune away (a deployment that only needs turn-taking can run the pure checkpoint). This is the one frontier that survives the diagnosis: the recall-versus-precision frontier dissolved when the labels became causal (§6), but the breadth-versus-precision cost at fixed adapter capacity does not, and it recedes with capacity, which is why the release uses rank 32; there the added replay and biasing schemas leave endpointing recall essentially intact, as they did not at rank 16. The context axis carries the same lesson: the spelled-entity counterfactual alone leaves natural-speech distractors hallucinated 6.3% of the time, and the natural-speech counterfactual folded into the released pool cuts this to 2.0%, below the base model's 3.7%, while preserving nearly all of the biasing uplift.

Costs. The endpoint fine-tune costs $+1.1/+2.7$ pp offline WER on LibriSpeech clean/other. Decode-time marker banning shows the markers are not the cause; the regression is narrow-schema drift, and the replay fraction in the released model recovers most of it (Appendix G, Table 12). Streaming WER, the deployment-relevant number, is unaffected. Serving runs on stock vLLM [Kwon et al., 2023] within real-time budgets through 16 concurrent sessions on a fractional-GPU slice, with bounded re-feed of the last audio window as the correct serving primitive and a force-flush bounding the WER and compute tails; details and caveats are in Appendix H, and the in-engine efficiency variant, windowed KV with the `<CTX>` prefix pinned as an attention sink [Xiao et al., 2024], is validated in a companion systems paper [Meng and Li, 2026].

6 Analysis

The claim that the labels are the load-bearing element of §3 rests on a controlled record: eight supervision compositions sharing architecture, adapter, data volume, and source corpora, differing only in the labels. This section presents the record, the intervention that diagnosed it, the mechanism behind the oscillation, and a synthetic study that isolates the mechanism outside speech.

6.1 Supervision-composition study

Table 4 compresses the record. The first composition supervised on offline clips alone (fire at every clip end); it scored perfectly offline and was unusable live, firing 89 times per speech-minute mid-utterance. Successive compositions added pools intended to fix this, and every addition moved the model *along* an apparent recall-versus-precision frontier, never off it. Two behaviors recurred across the family. Checkpoints were bimodal: eager ones fired mid-utterance, conservative ones missed turns, and no checkpoint did both jobs. And late compositions did

Table 4. The supervision-composition record. Eight compositions, one architecture, compressed into six rows: rows 1 and 2 each span two iterations (a data scale-up; a truncation-dose increase), and each later row adds one labeled pool to the mix above it, except row 6, which *replaces* row 5's same-audio hold pool with disjoint audio. "Early" latencies are negative: the model fires seconds before the turn ends. Row 7 is the causal recipe of §3.2.

#	Supervision change (cumulative)	Observed behavior
1	offline endpoint pool only: fire at every clip end	fires eagerly and "accurately" offline (100% fire recall); in streaming replay, fires 89× per speech-minute mid-utterance
2	+ truncated-speech hold pool (11%→38% of mix)	diminishing returns: median fire timing improves only $-13.1 \to -9.6 \to -7.6$ s (still seconds early) as the dose rises
3	+ trailing-silence fire pool	streaming latency crosses zero (P50 +0.32 s) but offline fire recall regresses 100→82%; the "trade-off" is first declared
4	+ complete utterances with *and* without silence tails, both labeled fire	silence becomes optional again; a strictly intermediate point on the same frontier; frontier declared fundamental
5	+ hold pool on the *same* audio as existing fire examples	collapse: opposite labels on identical inputs; the model stops firing on meeting audio entirely (fire recall 0%)
6	row 5's hold pool rebuilt on *disjoint but identically distributed* audio	bimodal oscillation between fire-mode and no-fire-mode attractors for the whole run (Figure 6, left); no checkpoint dominates; two-checkpoint ship recommended
7	**causal relabel (§3.2):** fire iff complete *and* $\geq$0.3 s observed silence	monotone training, both classes at ceiling simultaneously; single checkpoint dominates every predecessor (§5)

not converge at all: holdout accuracy oscillated for the entire run between a *fire-mode* attractor and a *no-fire-mode* attractor, as in Figure 6 (left). By the eighth composition (row 6) the working conclusion, recorded in the project log at the time, was a fundamental Pareto frontier, with a recommendation to ship two checkpoints, one eager and one conservative.

6.2 Diagnosis by intervention

The experiment. The frontier's fire-recall axis died by a one-variable intervention on the *evaluation*: we appended 1.0 s of silence to each clip of the legacy offline benchmark and re-scored the historical checkpoints (Table 5). The "broken" opposed-pools model's fire recall rose from 0.10 to 1.00 and the mixed-pools model's from 0.82 to 0.98, while the offline-labeled composition, which never depended on hearing silence, was the unchanged control. Under deployment-realistic input, all three fire essentially perfectly on complete utterances. The capability axis that four compositions of data engineering had been fighting was an artifact of where the evaluation clips ended.

The diagnosis. Why did the clips end where they did? Offline ASR datasets are segmented by forced alignment, so clips are cut at the end of speech, and evaluation clips inherit the cut. An offline endpoint benchmark built on such clips therefore demands a fire on audio that ends

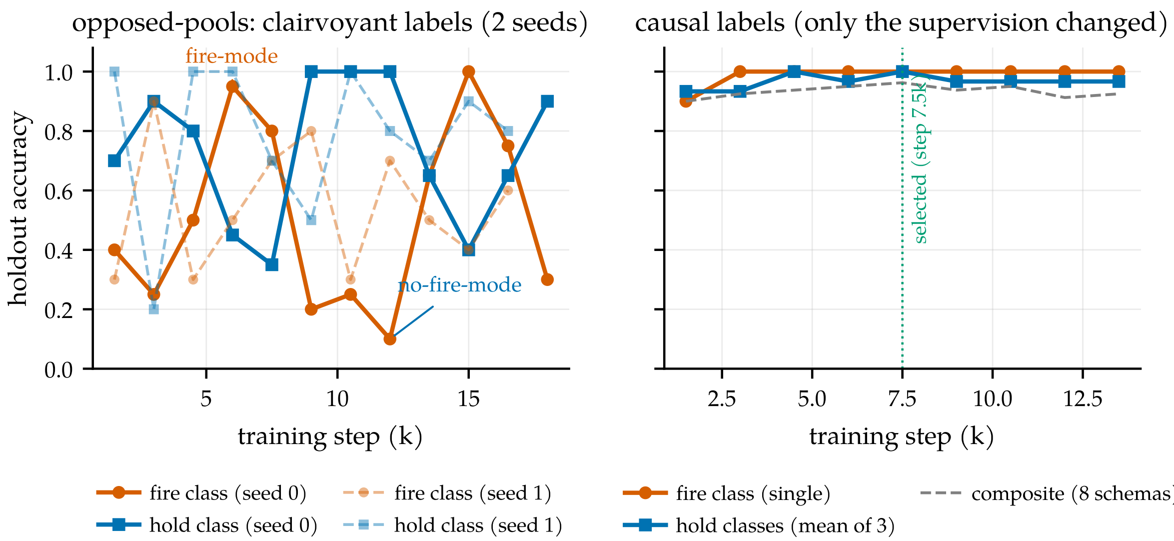

Figure 6. The fingerprint of contradictory supervision. Left: under the opposed-pools composition, holdout accuracy oscillates for the entire run between a fire-mode and a no-fire-mode attractor, reproducibly across two training seeds (solid, dashed); the oscillation is a property of the labels, not a seed. Right: the causal recipe of §3.2, with the same architecture, adapter, and data volume and only the labels changed, climbs monotonically, with both classes at ceiling simultaneously from step 3k on.

Table 5. The silence-append intervention. Fire recall on complete single utterances from the legacy offline benchmark, before and after appending 1.0 s of silence to each clip.

Model (supervision composition)	original clips	+1.0 s silence	Δ
offline-labeled (rows 1–2)	1.00	1.00	0.00
mixed-pools (row 3)	0.82	0.98	+0.16
opposed-pools (row 6)	0.10	1.00	+0.90

during or immediately after speech, a condition deployment never presents. The streaming pools taught the opposite, correctly: never fire before silence is observed. The condition "complete utterance, no trailing silence yet" thus carried the label *fire* in one pool and *hold* in the other; in the eighth composition this was 50/50 label noise on the exact decision the model exists to make. A second mechanism had the same structure: the historical pools separated "disfluent pause, hold" from "turn boundary, fire" using the transcript's segmentation. Measured at the decision point, mid-pause, the two classes are statistically indistinguishable: their silence-gap distributions overlap almost completely, and the disfluent pauses are the longer ones. The label is a function of who spoke *next*, that is, of the future. Both mechanisms are exactly the clairvoyance of §3.1, and nothing about either is specific to speech; Figure 11 (Appendix E) illustrates both leaks.

Why oscillation, specifically. Rows 5 and 6 of Table 4 form a natural two-condition experiment, differing only in where the contradictory hold labels sit. When opposite labels sit on literally identical audio (row 5), the conflict is visible to the loss on every example, and the model settles deterministically on one side: it simply stops firing. When opposite labels sit on disjoint but identically distributed audio (row 6), no single example exposes the conflict; each pool is separately learnable, incompatible only in aggregate, and training circulates between mode attractors that each satisfy one pool. The oscillation is not instability to be scheduled away; it is the loss surface reporting a contradiction that no individual example states. A logit-level probe (Appendix E, Figure 12) locates the circulation in the weights rather than at a decision threshold: marker-commit probability swings between snapshots in lockstep with the argmax, and no held-out example hovers at the decision boundary.

Evaluation is the port of entry. The legacy offline benchmark did not just mis-score the historical models; it steered four compositions of data engineering toward a frontier that does not exist, and the deployment-matched protocol of §4.1 re-*ranks* them (Appendix B, Figure 10): the offline champion is the worst live model, its offline strength being a premature-fire pathology, while the opposed-pools model, scored at 10% offline recall, is the best prior endpointer. Teams selecting turn-aware checkpoints on clip-cut offline evaluations should expect the same inversion.

6.3 Synthetic isolation of the mechanism

The failure reproduces outside speech, with no acoustics and no pretraining (setup in Appendix F). A two-layer, $\sim$30k-parameter transformer is trained on a synthetic stream in which completeness is readable from the prefix, each gap's outcome is not, and a tunable fraction f of the labels is clairvoyant by construction; at $f{=}0$ training is textbook monotone. As f grows, the fingerprint develops in order (Figure 13, Appendix F): fire- and hold-class accuracy become anti-correlated, checkpoints spread into an anti-diagonal cloud in the recall-precision plane (a phantom Pareto frontier that is in fact one model swinging, sampled at different steps), and true bistable mode-flipping onsets only at high fraction. The threshold is informative: the causal label remains the per-prefix *majority* for every $f{<}1$, so a single-mode optimum survives until the contradictory pool destabilizes it; the speech model of §6.1 lived in the high-f regime by construction, since near-balanced opposed pools are exactly what mixing offline supervision with streaming reality produces.

7 Discussion

A checklist. The failure mode generalizes to any model that must act mid-stream on offline-labeled data. Four questions for a streaming supervision pipeline:

Is your supervision clairvoyant?

1. **Prefix test.** For each label at decision time t: is it computable from input up to t? *(Our pause labels were keyed on the next speaker.)*
2. **Twin test.** Can two causally identical prefixes receive different labels anywhere in the data? *(Same complete-utterance prefix: fire in one pool, hold in the other.)* Minimal pairs differing in an *observable* are the fix, not the bug.
3. **Phantom-condition test.** Does the eval demand behavior on conditions deployment never produces? *(Clips cut at end-of-speech; deployment always delivers the next silence.)*
4. **Oscillation test.** Does training bounce between mode attractors that each satisfy

part of the data? That is the loss surface reporting a contradiction.

Limitations. Single-speaker, single-channel English; the mixed-channel meeting shape is future work. The dictation capability transfers zero-shot to shorter enumerations but not to longer-than-trained ones (on 16-digit card numbers the model fires at its learned ten-digit completeness point, transcription unaffected), so a learned completeness judge remains the general lever beyond pattern-specific schemas. The unified adapter's breadth-versus-precision and recall-versus-hallucination costs are quantified in §5.3, and a turn-taking-only deployment can run the pure endpointing checkpoint; concurrency numbers are contention-caveated lower bounds from a shared GPU (Appendix H).

8 Related work

Endpointing and end-of-turn detection. Classical endpointing thresholds a VAD [Silero Team, 2021] with a silence timeout [Raux and Eskenazi, 2008]; learned refinements predict end-of-query from acoustics [Shannon et al., 2017], jointly train an endpointer with the recognizer [Chang et al., 2019, Li et al., 2020, Bijwadia et al., 2023], or distill tiny acoustic end-of-turn models [Helwani et al., 2026, Pipecat team, 2025]. Recent systems fuse acoustic and linguistic cues [Wang et al., 2026, Yang et al., 2026] or fold VAD, turn detection, and ASR into one audio-LLM front-end [Li et al., 2026], architecturally closest to us but with no released weights or recipe. Deployed open detectors run beside the recognizer: Smart Turn (acoustic, open recipe, no transcript; Pipecat team, 2025) and the LiveKit and TEN end-of-utterance classifiers over a separate STT's output [LiveKit, 2025, TEN Team, 2025, LiveKit, 2026]. Open-weights recognizers with in-model turn signals, Kyutai STT's semantic-VAD head [Kyutai Labs, 2025] and NVIDIA's Parakeet-EOU token [NVIDIA, 2026], publish weights but not the labeling procedure or data construction, and neither measures context biasing. Commercial systems (Deepgram Flux, Deepgram, 2026; OpenAI semantic VAD, OpenAI, 2025; AssemblyAI, AssemblyAI, 2025) publish no training details. Our contribution is the missing artifact, the recipe, plus evidence that trade-offs this literature reports as intrinsic can be manufactured by non-causal supervision.

Turn-taking in dialogue, and other streaming decisions. Turn-taking prediction has a long tradition [Skantze, 2021], from text-based end-of-turn prediction [Ekstedt and Skantze, 2020] to voice-activity projection trained on future windows [Ekstedt and Skantze, 2022], with human timing evidence from Stivers et al. [2009], Levinson and Torreira [2015]. Projection objectives predict distributions over the future and are causally honest; our diagnosis concerns future-dependent quantities presented as deterministic labels. The same shape appears in simultaneous translation trained with reference-aligned read/write decisions [Ma et al., 2019] and in tool-call timing for agents cloned from completed trajectories, a streaming sibling of causal confusion in imitation learning [de Haan et al., 2019]; metric choices can similarly manufacture "emergence" [Schaeffer et al., 2023].

Streaming and unified ASR. Streaming recognition spans RNN-T [Graves, 2012], large weakly supervised models [Radford et al., 2023], decoder-only streaming with latency losses

[Wan et al., 2026], and unified streaming/offline speech-LLMs [Shi et al., 2026]; full-duplex speech-to-speech models [Défossez et al., 2024] dissolve the turn abstraction entirely. We inherit the unified-ASR substrate and add the in-transcript turn event plus the training recipe; vLLM's paged KV [Kwon et al., 2023] carries our serving path, with attention sinks [Xiao et al., 2024] and bounded-state serving [Meng and Li, 2026] as the in-engine efficiency variant of the pinned context prefix.

Context biasing. Contextual ASR ranges from attention-based deep context [Pundak et al., 2018] to retrieval-scale biasing [Gong et al., 2025], RL-tuned hotwords [Kong et al., 2025], and cue-based prompting [Novitasari et al., 2026]; Durmus et al. [2026] concurrently standardizes entity-level evaluation on our corpus; we contribute plain prompt-slot measurements on the open class.

9 Conclusion

Turn-aware streaming ASR has so far existed only behind commercial APIs. This paper makes it reproducible: one causal labeling rule, two pair constructions, and a deployment-matched benchmark turn a small open model into a single checkpoint that transcribes, endpoints semantically, handles dictation, and grounds transcription in context, at an operating point no silence timeout reaches. The obstacle was supervision, not architecture: offline corpora encode the future, and the resulting clairvoyant labels manufacture oscillation and phantom trade-offs, diagnosed by intervention and cured, twice, by counterfactual construction. The failure class, and its checklist diagnostic, should transfer to any streaming decision trained from offline logs.

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References

AssemblyAI. Universal-streaming: speech-to-text with intelligent endpointing. Product documentation, `https://www.assemblyai.com/docs`, 2025.

Shaan Bijwadia, Shuo-Yiin Chang, Bo Li, Tara N. Sainath, Chao Zhang, and Yanzhang He. Unified end-to-end speech recognition and endpointing for fast and efficient speech systems. In *IEEE SLT*, 2023.

Jean Carletta et al. The AMI meeting corpus: A pre-announcement. In *International Workshop on Machine Learning for Multimodal Interaction*, 2005.

Shuo-Yiin Chang, Rohit Prabhavalkar, Yanzhang He, Tara N. Sainath, and Gabor Simko. Joint endpointing and decoding with end-to-end models. In *ICASSP*, 2019.

Pim de Haan, Dinesh Jayaraman, and Sergey Levine. Causal confusion in imitation learning. In *NeurIPS*, 2019.

Deepgram. Flux: Conversational speech recognition with model-integrated end-of-turn detection. Product documentation, https://deepgram.com, 2026.

Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. Moshi: a speech-text foundation model for real-time dialogue. *arXiv preprint arXiv:2410.00037*, 2024.

Miguel Del Rio, Peter Ha, Quinten McNamara, Corey Miller, and Shipra Chandra. Earnings-22: A practical benchmark for accents in the wild. *arXiv preprint arXiv:2203.15591*, 2022.

Berkin Durmus, Chen Cen, Eduardo Pacheco, Arda Okan, and Atila Orhon. Contextual earnings-22: A speech recognition benchmark with custom vocabulary in the wild. arXiv preprint arXiv:2604.07354, 2026.

Erik Ekstedt and Gabriel Skantze. TurnGPT: a transformer-based language model for predicting turn-taking in spoken dialog. In *Findings of EMNLP*, 2020.

Erik Ekstedt and Gabriel Skantze. Voice activity projection: Self-supervised learning of turn-taking events. In *Interspeech*, 2022.

Xun Gong, Anqi Lv, Zhiming Wang, Huijia Zhu, and Yanmin Qian. BR-ASR: Efficient and scalable bias retrieval framework for contextual biasing ASR in speech LLM. In *Interspeech*, 2025. arXiv:2505.19179.

Alex Graves. Sequence transduction with recurrent neural networks. *arXiv preprint arXiv:1211.3711*, 2012.

Karim Helwani, Hoang Do, James Luan, and Sriram Srinivasan. A hierarchical end-of-turn model with primary speaker segmentation for real-time conversational AI. *arXiv preprint arXiv:2603.13379*, 2026.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *ICLR*, 2022.

Zohar Jackson, César Souza, Jason Flaks, Yuxin Pan, Hereman Nicolás, and Adhish Thite. Jakobovski/free-spoken-digit-dataset: v1.0.8, 2018. URL https://doi.org/10.5281/zenodo.1342401.

Keller Jordan, Yuchen Jin, Vlado Boza, Jiacheng You, Franz Cesista, Laker Newhouse, and Jeremy Bernstein. Muon: An optimizer for hidden layers in neural networks. https://kellerjordan.github.io/posts/muon/, 2024.

YuXiang Kong, JunFeng Hou, Jian Tang, Bingqing Zhu, Jicheng Zhang, and Shaofei Xue. Contextual biasing for LLM-based ASR with hotword retrieval and reinforcement learning. *arXiv preprint arXiv:2512.21828*, 2025.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *SOSP*, 2023.

Kyutai Labs. Kyutai STT: Streaming speech-to-text with delayed streams modeling. https://github.com/kyutai-labs/delayed-streams-modeling, 2025.

Stephen C. Levinson and Francisco Torreira. Timing in turn-taking and its implications for processing models of language. *Frontiers in Psychology*, 6:731, 2015.

Bo Li, Shuo-Yiin Chang, Tara N. Sainath, Ruoming Pang, Yanzhang He, Trevor Strohman, and Yonghui Wu. Towards fast and accurate streaming end-to-end asr. In *ICASSP*, 2020.

Bojie Li. User as code: Executable memory for personalized agents, 2026. arXiv:2606.16707.

Yadong Li, Guoxin Wu, Haiping Hou, and Biye Li. UAF: A unified audio front-end LLM for full-duplex speech interaction. *arXiv preprint arXiv:2604.19221*, 2026.

LiveKit. Improving voice AI's turn detection with transformers. https://github.com/livekit/agents, model livekit/turn-detector, 2025.

LiveKit. Improved end-of-turn model cuts voice AI interruptions 39%. Blog post, https://livekit.com/blog/improved-end-of-turn-model-cuts-voice-ai-interruptions-39. Accessed 2026-07-05, 2026.

Mingbo Ma, Liang Huang, Hao Xiong, Renjie Zheng, Kaibo Liu, Baigong Zheng, Chuanqiang Zhang, Zhongjun He, Hairong Liu, Xing Li, Hua Wu, and Haifeng Wang. STACL: Simultaneous translation with implicit anticipation and controllable latency using prefix-to-prefix framework. In *ACL*, 2019.

Jiaying Meng and Bojie Li. Metronome: Bound the cache, keep the beat for real-time interaction model serving, 2026. arXiv:2607.02640.

Sashi Novitasari, Takashi Fukuda, Kurata Gakuto, and George Saon. Contextual biasing for ASR in speech LLM with common word cues and bias word position prediction. *arXiv preprint arXiv:2604.12398*, 2026.

NVIDIA. Parakeet-realtime-eou-120m: streaming ASR with end-of-utterance prediction. Model card, https://huggingface.co/nvidia/parakeet_realtime_eou_120m-v1. Accessed 2026-07-05, 2026.

OpenAI. Semantic VAD in the realtime API. Product documentation, https://platform.openai.com/docs, 2025.

Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. Librispeech: an ASR corpus based on public domain audio books. In *ICASSP*, 2015.

Pine AI. Pine AI: The most natural human-computer interface is your voice. Blog post, 2026. URL https://www.19pine.ai/blog/pine-ai-the-most-natural-human-computer-interface-is-your-voice. Accessed 2026-06-28.

Pipecat team. Smart turn: an open turn-detection model. https://github.com/pipecat-ai/smart-turn, 2025.

Golan Pundak, Tara N. Sainath, Rohit Prabhavalkar, Anjuli Kannan, and Ding Zhao. Deep context: End-to-end contextual speech recognition. In *IEEE SLT*, 2018.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *ICML*, 2023.

Antoine Raux and Maxine Eskenazi. Optimizing endpointing thresholds using dialogue features in a spoken dialogue system. In *Proceedings of SIGdial*, 2008.

Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? In *NeurIPS*, 2023.

Matt Shannon, Gabor Simko, Shuo-Yiin Chang, and Carolina Parada. Improved end-of-query detection for streaming speech recognition. In *Interspeech*, 2017.

Xian Shi, Xiong Wang, Zhifang Guo, Yongqi Wang, Pei Zhang, Xinyu Zhang, Zishan Guo, Hongkun Hao, Yu Xi, Baosong Yang, Jin Xu, Jingren Zhou, and Junyang Lin. Qwen3-ASR technical report. *arXiv preprint arXiv:2601.21337*, 2026.

Silero Team. Silero VAD: pre-trained enterprise-grade voice activity detector. https://github.com/snakers4/silero-vad, 2021.

Gabriel Skantze. Turn-taking in conversational systems and human-robot interaction: A review. *Computer Speech & Language*, 67, 2021.

Tanya Stivers, Nicholas J. Enfield, Penelope Brown, Christina Englert, Makoto Hayashi, Trine Heinemann, Gertie Hoymann, Federico Rossano, Jan Peter de Ruiter, Kyung-Eun Yoon, and Stephen C. Levinson. Universals and cultural variation in turn-taking in conversation. *Proceedings of the National Academy of Sciences*, 106(26):10587–10592, 2009.

TEN Team. TEN turn detection: turn detection for full-duplex dialogue communication. https://github.com/TEN-framework/ten-turn-detection, 2025.

Genshun Wan, Wenhui Zhang, Jing-Xuan Zhang, Shifu Xiong, Jianqing Gao, and Zhongfu Ye. Streaming speech recognition with decoder-only large language models and latency optimization. *arXiv preprint arXiv:2601.22779*, 2026.

Chengyou Wang, Hongfei Xue, Chunjiang He, Jingbin Hu, Shuiyuan Wang, Bo Wu, Yuyu Ji, Jimeng Zheng, Ruofei Chen, Zhou Zhu, and Lei Xie. FastTurn: Unifying acoustic and streaming semantic cues for low-latency and robust turn detection. *arXiv preprint arXiv:2604.01897*, 2026.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. In *ICLR*, 2024.

Guangzhao Yang, Yu Pan, Shi Qiu, and Ningjie Bai. JAL-Turn: Joint acoustic-linguistic modeling for real-time and robust turn-taking detection in full-duplex spoken dialogue systems. *arXiv preprint arXiv:2603.26515*, 2026.

A Label-specification details

Completeness classifier for pair examples (Table 1, #4–5): utterance A is *incomplete* iff its transcript ends in a filler/connective ("and, but, so, or, uh, um, the, a, to, of, in, with, i mean, you know, . . . "), ends mid-word (AMI partial-word annotation), or has fewer than 3 words outside a closed acknowledgment list. Borderline cases land in the resume-after-fire cost metric (§4.1); they never create fire/hold contradictions on identical observables. Abandonment (incomplete speech, speaker never resumes) is an inference-policy timeout, not a label. Pair examples use both same- and different-speaker AMI pairs: the fire decision keys on completeness + silence, never on speaker identity, which single-channel deployment cannot observe. Figure 8 illustrates the minimal-pair construction of §3.2, and Figure 7 places it beside its contextual analog, the counterfactual twin of §3.3.

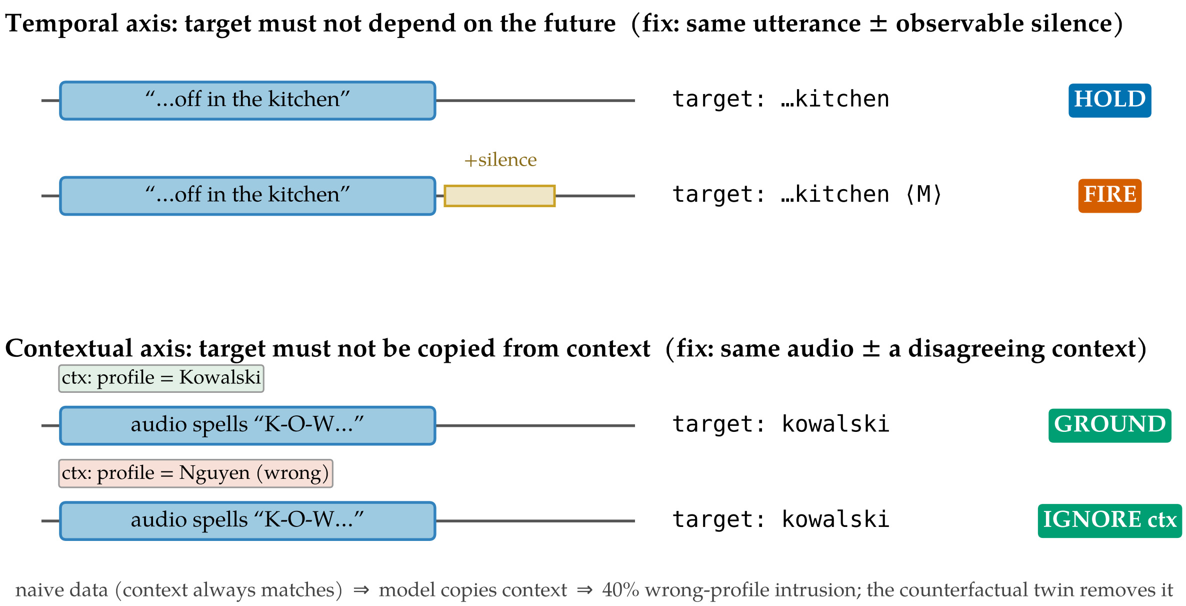

Figure 7. One failure class, two instances. Temporal axis: labels that peek at the future manufacture a phantom frontier; the fix is the minimal pair, the same utterance with and without an observable silence tail. Contextual axis: a context prefix that always matches the audio becomes a copyable shortcut; the fix is the counterfactual twin, the same audio with a matching and a disagreeing context, target following the audio.

B Benchmark specification and diagnostics

Full protocol: 0.5 s chunks; committed-prefix incremental decoding (emitted text committed, last 5 tokens rolled back) matching the base model's official streaming semantics; fires timestamped at the chunk boundary where <END_SPEECH> first appears; recall window $[-0.25, +1.5]$ s clipped at the next utterance's onset. Set sizes: development 25 stretches (96 turn-final boundaries), post-development confirmation 50 stretches (184 boundaries), held-out 100 stretches (384 boundaries). The first version of the boundary classifier promoted a boundary to turn-final when another meeting speaker interleaved before the target speaker resumed. On a single-channel stream that speech is silence; demanding a fire there is exactly the clairvoyance the benchmark exists to remove. The corrected classification moved the mixed-pools model's

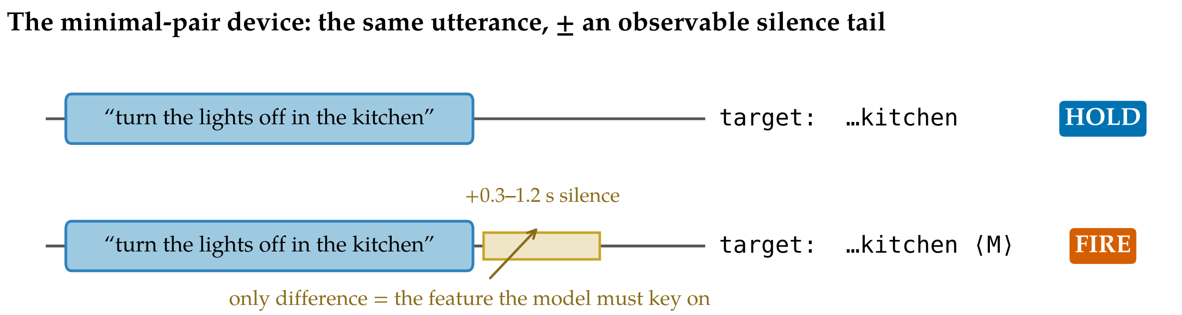

Figure 8. The minimal-pair device. Each complete utterance appears twice: without a silence tail (target: transcript only) and with a 0.3–1.2 s tail (target: transcript + marker). The only difference between the opposite-labeled examples is the observable the model must key on.

recall 0.917→0.906 and the opposed-pools model's 0.935→0.938; no conclusion changed, but the episode is a working example of the §7 checklist applied to one's own evaluation.

Silence incompetence, quantified. Models trained only on speech-initial examples hallucinate on silence: spurious fires of the ungated mixed-pools model track each stretch's silence content almost perfectly (r=0.997; Figure 9), at ∼1.9 fires per second of silence. This is why §4.1 runs every comparison behind the energy gate.

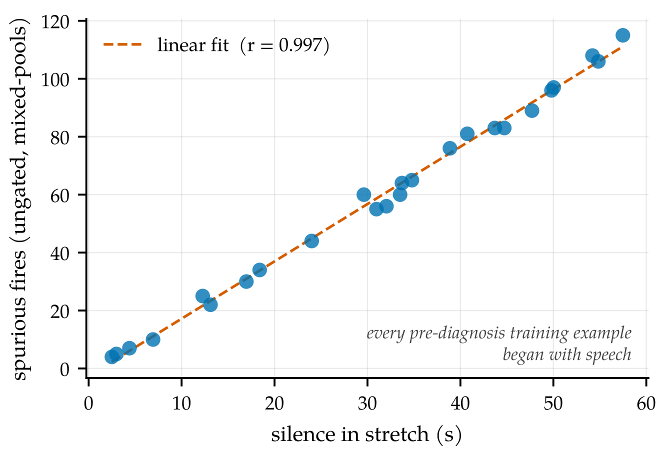

Figure 9. Silence incompetence. Spurious fires of the ungated mixed-pools model track the silence content of each stretch almost perfectly; ungated recall is therefore uninterpretable, and all reported arms run behind the energy gate.

The ranking inversion. Figure 10 shows the historical model ranking on the legacy offline benchmark versus the deployment-matched replay benchmark (§6.2): the offline champion fires 89 times per speech-minute once audio keeps flowing, while the opposed-pools model, written off at 10% offline recall, is the best prior endpointer.

C The turn-aware system landscape

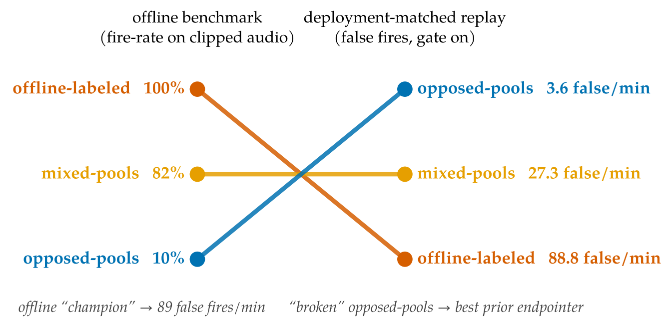

Figure 10. The ranking inversion. Model ranking on the legacy offline benchmark (left) versus the deployment-matched replay benchmark (right). The offline champion is the worst live model.

Table 6. The landscape (verified July 2026; discussion in §8). "Turn detectors" are separate models beside a recognizer; "cascade" is VAD + silence timeout + separate ASR, the deployed default and our measured baseline.

	This work	Flux / `sem.vad`	Kyutai STT	Parakeet-EOU	Turn det.	Cascade
Streaming ASR	✓	✓	✓	✓	–	✓
In-model end-of-turn	semantic	semantic	sem. head	token	sep. model	timeout
Open weights	✓	–	✓	✓	mostly	✓
Turn-training recipe	✓	–	–	–	Smart Turn	n/a
Context biasing, measured	✓ (+28.9 pp)	–	–	–	–	ASR-dep.

D Additional results

Gated arms and the confirm-horizon sweep. Table 7. The confirm horizon behaves as a clean latency-vs-false-fire dial: $h=1$ strictly improves the mixed-pools model (recall *rises* because deferring the fire lets the segment gather the boundary inside tolerance) and zeroes the opposed-pools and causal models' false fires; $h=2$ over-suppresses (mixed-pools recall 0.812, latency past budget). The causal model is the only checkpoint whose $h=0$ row is already deployable. Ungated arms are omitted as uninterpretable (Figure 9): with $\sim$1.9 spurious fires per second on silence, the pre-diagnosis models score 0.95–0.99 "recall" by chance.

Table 7. All gated arms on the 25-stretch development benchmark. Resume = resume-after-fire rate over continuation boundaries (a cost metric, not an error rate). WER is a fraction, unbounded above through insertions.

Arm	Recall	P50	P95	False/min	Resume	WER_{mean}
offline-labeled + gate	0.896	0.05 s	0.25 s	88.8	0.89	0.85
mixed-pools + gate	0.906	0.16 s	0.89 s	27.3	0.96	0.36
mixed-pools + gate + confirm $h=1$	0.927	0.68 s	1.27 s	3.2	0.85	0.36
mixed-pools + gate + confirm $h=2$	0.812	1.15 s	1.42 s	1.1	0.56	0.36
opposed-pools + gate	0.938	0.26 s	0.56 s	3.6	0.93	4.96
opposed-pools + gate + confirm $h=1$	0.938	0.77 s	1.06 s	0.0	0.85	4.96
causal + gate	0.969	0.39 s	0.70 s	0.3	1.00	1.43
causal + gate + confirm $h=1$	0.969	0.89 s	1.20 s	0.0	0.93	1.43

The full timeout family. Table 8, both detectors on the development (25 stretches) and fresh confirmation (50 stretches) sets. The family's shape is stable across sets: recall collapses between X=1.0 and 1.5 s (the fire lands outside the +1.5 s tolerance), and no setting reaches the causal model's (recall, latency, false-fire) point; matching its recall costs a timeout 3× the latency and ∼5× the false fires. Silero [Silero Team, 2021] ran per-chunk with state resets, a slight handicap; RMS shares the LM arms' exact detector and is the apples-to-apples policy comparison (on this close-talk channel the Silero cascade is dominated by plain RMS).

Table 8. Silence-timeout cascade, all settings, on the development (dev) and fresh confirmation (conf) stretch sets.

Detector	Set	X (s)	Recall	P50	P95	False/min	Resume
RMS	dev	0.5	0.958	0.65 s	0.94 s	6.3	1.00
RMS	dev	1.0	0.969	1.15 s	1.44 s	1.4	0.81
RMS	dev	1.5	0.208	1.38 s	1.49 s	0.8	0.07
RMS	dev	2.0	0.031	0.55 s	0.55 s	0.2	0.00
RMS	conf	0.5	0.978	0.70 s	0.95 s	4.6	0.99
RMS	conf	1.0	0.989	1.20 s	1.45 s	0.6	0.73
RMS	conf	1.5	0.185	1.44 s	1.50 s	0.4	0.07
RMS	conf	2.0	0.016	1.14 s	1.14 s	0.2	0.00
Silero	dev	0.5	0.844	0.54 s	0.88 s	13.5	0.96
Silero	dev	1.0	0.854	1.04 s	1.38 s	5.5	0.85
Silero	dev	1.5	0.417	1.24 s	1.49 s	2.5	0.41
Silero	dev	2.0	0.125	1.32 s	1.42 s	1.0	0.00
Silero	conf	0.5	0.837	0.51 s	0.83 s	12.3	0.88
Silero	conf	1.0	0.864	1.00 s	1.33 s	4.4	0.73
Silero	conf	1.5	0.451	1.29 s	1.48 s	2.0	0.27
Silero	conf	2.0	0.130	1.34 s	1.44 s	1.3	0.07

Fresh-set confirmation. The causal model with the gate on the confirmation set: recall 0.924, P50 0.42 s, P95 0.70 s, 0.21 false fires per speech-minute, resume 0.91, WER median 0.30 / mean 4.63 (three long-monologue stretches enter the no-flush repetition loop; the force-flush of Appendix H is the documented fix). The 4.5 pp recall drop from the development set (93/96 vs. 170/184 turn-final boundaries) is consistent with sampling variation: the two-proportion 95% interval on the difference, $[-0.7, +9.7]$ pp, includes zero.

Probe confidence intervals and the dictation scoring window. On the enlarged probes (250 dictation sequences; 240 spelled items over 120 identities), the released model's 95% intervals are: premature fires per sequence 0.16 $[0.12, 0.22]$, final recall 0.88 $[0.84, 0.92]$, email-with-context 0.93 $[0.87, 0.97]$, intrusion 0.008 $[0.002, 0.030]$; and on the 100-stretch replay set, boundary recall 0.982 $[0.965, 0.995]$ at 1.30 $[0.91, 1.71]$ false fires per speech-minute. The shipped scoring window books any fire within +0.25 s of the last speech end as premature and denies it final-recall credit. Re-scoring every fire by its offset δ from the true end of speech (genuine mid-sequence premature: $\delta < -0.30$ s; on-time: $-0.30 \leq \delta \leq +1.75$ s; late: $\delta > +1.75$ s) gives, on the same 250 sequences: 0.07 genuine mid-sequence fires per number (18/250 sequences), 0.968 on-time final recall, zero late fires, and end-fire latency P50 0.44 s (fires quantized up to the 0.5 s chunk grid). The residual under the shipped window is therefore eagerness within one chunk of the true end, not lateness; we report the shipped window in Table 3 for comparability across rows and note that it is conservative for a low-latency endpointer.

Released-model endpointing across sets. Table 9 collects the released rank-32 checkpoint's replay numbers, quoted piecewise in §5.3, next to the pure-endpointing causal checkpoint of §5.1. The breadth-versus-precision cost (0.97 versus 0.3 false fires per speech-minute) is a development-set comparison at matched latency; a one-chunk confirm horizon zeroes the released model's false fires at 0.958 recall (P50 0.89 s) for deployments that prefer conservatism over latency, and the added schemas leave endpointing recall essentially intact at rank 32 (0.95 versus 0.97 on the development set).

Table 9. The released unified checkpoint (rank-32) on the replay benchmark, both stretch sets, alongside the pure-endpointing causal checkpoint of §5.1. dev = 25-stretch development set; held-out 100 = the 100-stretch set; fresh 50 = the post-development confirmation set.

Model	Set	Recall	P50	P95	False/min
released r32 + gate	dev	0.948	0.39 s	0.65 s	0.97
released r32 + gate + confirm h=1	dev	0.958	0.89 s	1.15 s	0.0
released r32 + gate	held-out 100	0.982	0.38 s	0.64 s	1.30
pure endpoint (causal, r16) + gate	dev	0.969	0.39 s	0.70 s	0.3
pure endpoint (causal, r16) + gate	fresh 50	0.924	0.42 s	0.70 s	0.2

Seed robustness of the released recipe. Table 10. Retraining the released recipe (rank-32 adapter, unified ~20k pool with 20% plain-ASR replay and natural-speech biasing, weight-decay and cosine fixes) under three seeds and selecting each best checkpoint on the probe axes gives stable premature-fire, context, and intrusion numbers; the unified capability is a property of the recipe, not a lucky checkpoint.

Table 10. Released recipe under three training seeds (best checkpoint each, smaller common probe set; the released checkpoint's enlarged-probe figures, email 0.93 and intrusion 0.8%, are in Table 3). Seed 0 is the released checkpoint.

Seed	Premature/seq $\downarrow$	Email +ctx $\uparrow$	Intrusion $\downarrow$
0 (released)	0.16	0.88	0.000
1	0.16	0.98	0.000
2	0.16	0.85	0.000
mean $\pm$ sd	0.16 $\pm$ 0.00	0.90 $\pm$ 0.07	0.000 $\pm$ 0.000

Schema ablation. Table 11. We retrain the endpointing recipe on the full endpointing pool with one construction removed at a time (iso-count: the dropped quota is redistributed across the remaining schemas, holding volume fixed), select each arm on the holdout composite, and score on the 25-stretch replay set. Each construction has a distinct predicted failure; two of three reproduce it. Removing the **minimal pair** (schema 2) makes silence optional again: gated false fires rise 1.08→5.28 per speech-minute and the confirm horizon cancels 34 candidates versus 10, so the pathology and its inference crutch both return. Removing the **silence schemas**

(7–8) costs silence-competence: ungated, the model emits 1458 markers on silence versus 8, masked in the gated numbers by the energy gate. Removing the **pause pair**'s incomplete half (schema 5) does not raise false fires; the model is mildly more conservative (recall 0.948→0.906, false fires 1.08→0.43), so pause discrimination is largely covered by schema 4's complete-A gaps.

Table 11. Schema ablation on the endpointing pool (iso-count; 25-stretch replay, best checkpoint each). Each arm is an independent retrain selected on the holdout composite, so the baseline here (0.948/1.08) is a different checkpoint from the causal model of Table 2 (0.969/0.3). "Silence-fires" is the ungated marker count on silence; "h=1 cancels" is the confirm-horizon crutch metric.

Arm	Recall ↑	False/min ↓	Silence-fires ↓	h=1 cancels	Predicted failure?
baseline (full pool)	0.948	1.08	8	10	–
−minimal pair (#2)	0.906	5.28	5	34	yes (silence optional)
−silence (#7–8)	0.969	1.94	1458	12	yes (silence spray)
−pause pair (#5)	0.906	0.43	8	4	no (redundant)

Offline WER regression. Table 12. The fine-tune costs +1.1/+2.7 pp on the full official LibriSpeech splits (single-shot decode on vLLM, jiwer/whisper normalization). "Fires" is the fraction of clips where the model emitted the endpoint marker; offline clips end at or near end-of-speech, and the base model never fires because its marker rows are untrained. Banning the markers at decode time zeroes the fires without moving WER, which rules out decode truncation as the cause; Appendix G attributes the regression to narrow-schema drift and quantifies the plain-ASR-replay mitigation. The released unified model, which folds a 20% plain-ASR-replay fraction into a rank-32 adapter (§3.3), recovers most of the gap, scoring 3.5%/6.9%.

Table 12. Offline WER (LibriSpeech, full official splits).

	WER ↓		Fires	
Model	clean	other	clean	other
Qwen3-ASR-0.6B (base)	2.79%	5.14%	0%	0%
+ endpoint LoRA (merged, causal, endpoint-only)	3.93%	7.87%	4.7%	5.9%
+ markers banned at decode	3.94%	7.91%	0%	0%
released unified (r32, +replay)	**3.49%**	**6.90%**	3.2%	4.7%

E Diagnosis details

The two leaks, illustrated. Figure 11 shows the two mechanisms of §6.2 side by side: the forced-alignment clip cut that puts opposite labels on the same prefix, and the next-speaker pause labels whose classes are indistinguishable at the decision point (gap medians 1.39 s for disfluent pauses vs. 0.83 s for turn boundaries).

The logit-level oscillation probe. Figure 12 is the measurement behind the mode-circulation claim of §6. The trained marker pathway is a rigid two-token sequence (`<EAGER_END_SPEECH>`

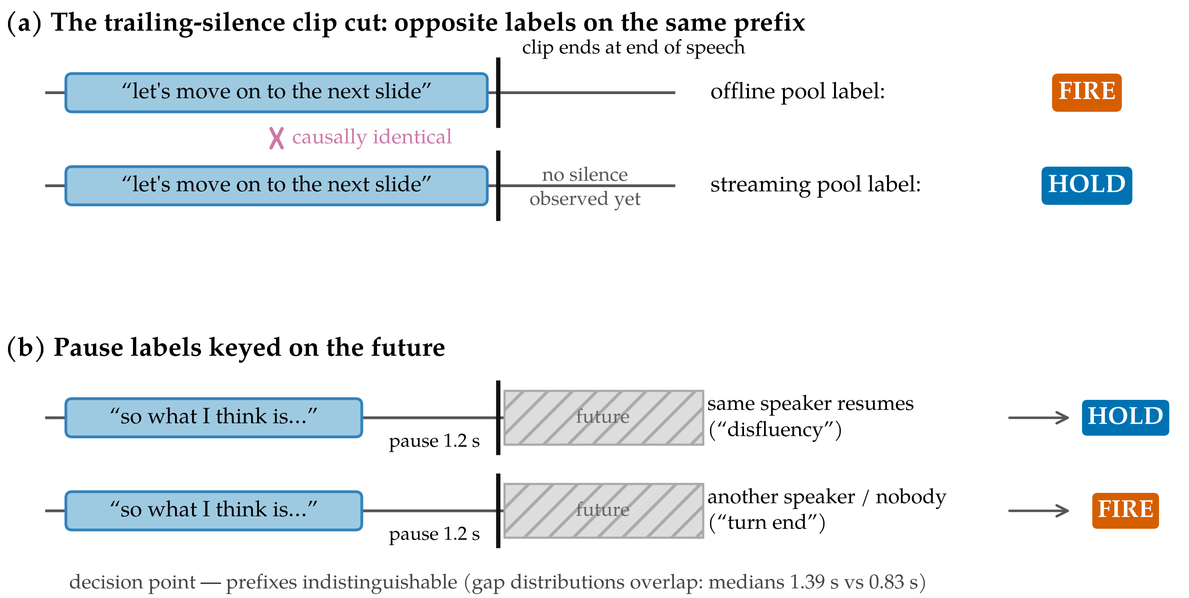

Figure 11. Two ways offline-derived supervision leaks the future. (a) Offline clips inherit forced-alignment cuts at end of speech, so the offline pool demands a fire on exactly the prefix the streaming pool labels *hold*. (b) Disfluency-vs-boundary pause labels are keyed on who speaks next; at the decision point the two classes' silence-gap distributions overlap almost completely.

then <END_SPEECH>) whose internal bigram is deterministic once entered, so we define P_{fire} as the marginal probability that the continuation enters the marker path rather than terminating, evaluated at the end-of-audio decision position with the transcript teacher-forced (marginalizing over the first continuation token at $\geq$0.98 probability coverage and greedy-rolling each branch). On the balanced holdout of the opposed-pools run, the exact examples behind Figure 6 (left), mean P_{fire} tracks the recorded argmax accuracy (r=+0.98 on the contested fire schema; r=−0.95 against hold-class accuracy) and swings with amplitude $\sim$0.4 across snapshots. The per-example distribution rules out threshold flapping: not one of the 40 held-out examples sits in the ambiguous $[0.2, 0.8]$ band across all eleven snapshots (only 13–15% sit there at any single snapshot), and 7 of 40 fully reverse between confident fire ($>$0.8) and confident hold ($<$0.2) as training proceeds. The trailing-silence schema, on which both pools agree, stays pinned at $P_{\text{fire}}\approx 1.0$ throughout: the circulation is confined to the schemas the pools contest.

F The synthetic clairvoyant-fraction experiment

Setup for the §6.3 experiment: a discrete stream of token phrases separated by silence runs, with no acoustics and no pretraining. A phrase ends complete or incomplete, always readable from the prefix, while each gap's full length (speaker resumes vs. turn ends, 50/50) is knowable only after the decision point. At a fixed depth into every gap, a two-layer causal transformer ($\sim$30k parameters) must emit fire or hold. The *causal* rule fires iff the phrase was complete; the *clairvoyant* rule fires iff the gap turns out long, a genuine future dependence and the direct analog of clips cut at boundaries and transcripts recording who spoke next. Completeness and gap length are independent, so the two rules are maximally opposed while each remains internally consistent, reproducing the "internally consistent pools with incompatible optima"

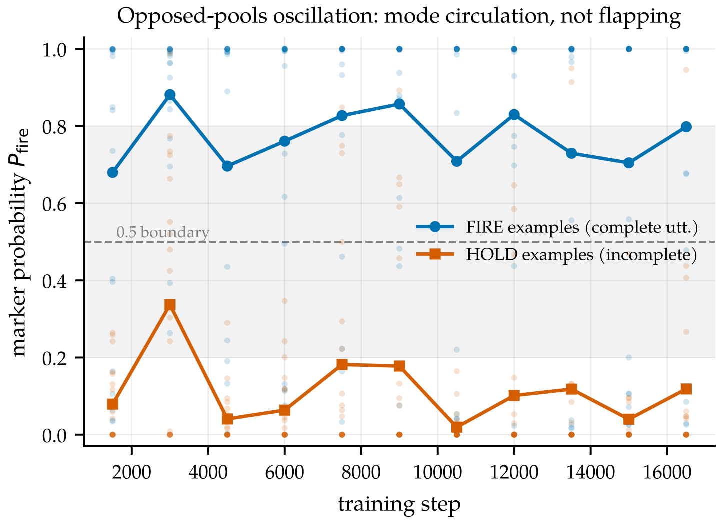

Figure 12. Mode circulation, not threshold flapping. Mean marker-commit probability P_{fire} for fire-labeled (complete utterance) and hold-labeled (incomplete) holdout examples across the opposed-pools run; small dots are per-example values. The two classes swing in lockstep as the weights orbit between the fire-mode and no-fire-mode attractors; no example hovers persistently at the 0.5 decision boundary.

structure of §3.1. A fraction f of training sequences is clairvoyantly labeled; the holdout is always causal; three seeds per f; everything runs on CPU in minutes.

The dose-response in detail (Figure 13): at f=0, both classes reach 1.00, zero mode flips, worst post-warmup holdout accuracy 0.99. The fire/hold accuracy anti-correlation grows monotonically (-0.3 at f=0.1, -0.5 at f=0.5, -1.0 at f=1.0), while true bistable mode-flipping onsets only at high fraction: 8 flips per run at f=0.85 and 16 at f=1.0, zero at $f\leq0.7$.

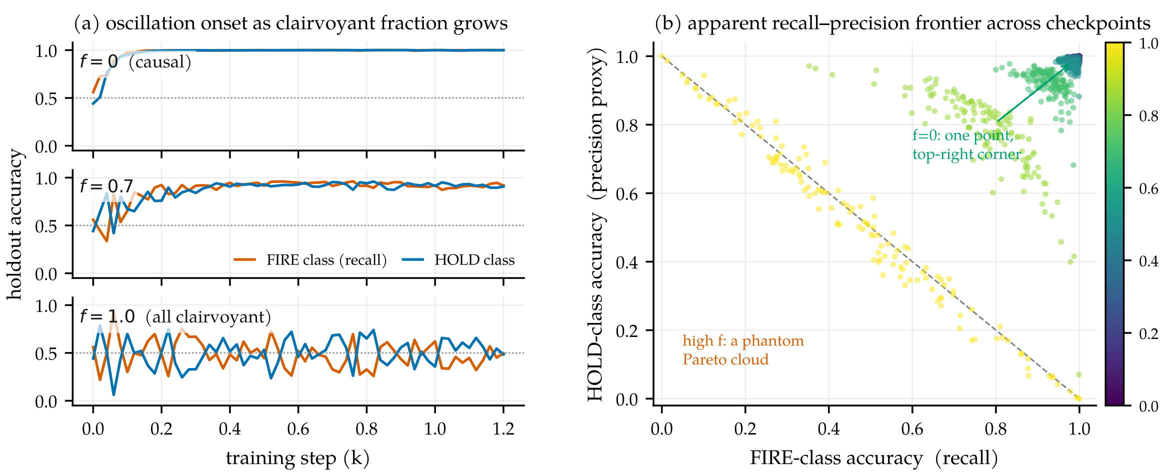

Figure 13. The mechanism, isolated. A ∼30k-parameter transformer on a synthetic stream with a tunable clairvoyant-label fraction f. (a) Holdout trajectories: monotone convergence at f=0, visible noise at f=0.7, bistable mode-oscillation at f=1.0. (b) Every checkpoint's (fire recall, hold precision), colored by f: the causal run collapses to one top-right point; high f spreads checkpoints into an anti-diagonal cloud, a phantom Pareto frontier traced by one oscillating model.

G Training-recipe notes

All models share one adapter configuration: LoRA on the attention q/k/v/o projections plus the two marker embedding rows, $\alpha=2r$, batch 8 ($r=16$ for the endpointing experiments, $r=32$ for the released unified checkpoint, which is selected on the probe axes of dictation, context, and intrusion rather than the holdout composite alone). The unified model (§3.3) adds two AdamW hygiene fixes and one negative result. All three were isolated on identical data (the unified pool), so the comparisons below vary only the optimizer setting.

A weight-decay bug worth fixing, but not the WER culprit. We reserve two vocabulary rows as markers and train them by unfreezing the (tied) token-embedding and output-head matrix and masking gradients to those two rows. But AdamW's *decoupled* weight decay applies $p \leftarrow p - \eta\lambda p$ to *every* row regardless of gradient, so at $\eta=2\times10^{-4}$, $\lambda=0.01$ the entire (tied) 151k-row matrix shrinks $\sim$2.4% over 12k steps, silently degrading the base model's vocabulary and output head. This is a general hazard for any recipe that adds a few trainable vocabulary rows to a frozen embedding, and we fix it with a $\lambda=0$ group. The negative half: in a controlled comparison on identical data, the fix (plus cosine schedule) did *not* lower offline WER; the fixed unified checkpoint scores 3.9%/7.3% (clean/other) versus 3.6%/7.0% without it, within checkpoint-selection noise and in the wrong direction. The offline-WER regression is therefore *not* caused by the weight-decay bug; it is narrow-schema drift from fine-tuning on $\sim$12k examples with no plain-ASR replay data. We keep the fix for its base-model-preservation rationale, not for a WER win we cannot demonstrate.

Plain-ASR replay: the fix that does move WER. The released model mixes $\sim$4k plain transcription examples (audio $\rightarrow$ transcript, no marker, no context) into the $\sim$20k-example pool, a 20% fraction. They teach the model that a complete utterance with no context is simply to be transcribed, recovering offline WER from 3.9%/7.3% (recipe fixes, no replay) to 3.5%/6.9% and cutting spurious offline marker fires from $\sim$10–20% to 3.2%. The replay fraction trades against endpointing sharpness at fixed adapter capacity; raising the LoRA rank from 16 to 32 buys back that sharpness, which is why the released checkpoint is rank-32 (§5.3).

Cosine decay. A cosine schedule from 2×10^{-4} to 10^{-5} after a 100-step warmup gives the most stable, highest holdout-composite trajectory (plateau 0.992 from step 6k) and a slightly less trigger-happy final checkpoint than the constant schedule.

Muon underperforms AdamW here. We tried Muon [Jordan et al., 2024] on the 2-D LoRA factors (Newton-Schulz-orthogonalized momentum, AdamW retained for the marker rows) at a 100$\times$ larger base LR (2×10^{-2}, since the orthogonalized update is $\sim$unit-norm). On identical data Muon's raw loss descends *faster* early but its task metric plateaus at holdout composite 0.846 (by step 3–4.5k, conversational schema collapsing) versus AdamW's 0.985. The cause is structural: rank-16 LoRA factors (1024×16) are too rectangular for the 5-step Newton-Schulz iteration to orthogonalize (empirically $\|O^\top O - I\|$ off-diagonals $\approx$0.5), and orthogonalizing the two factors separately is not the same operation as orthogonalizing their product. We keep AdamW.

H Serving details

Three findings from running the model on shared-GPU vLLM [Kwon et al., 2023] rather than a simulator (Figure 14), summarized in §5.3. The natural flat-cost serving shape, appending each 0.5 s chunk as a new audio placeholder in a resident request, is out of distribution for a model trained on single segments and destroys it (88% WER); bounded re-feed of the last window is the correct primitive (3.8%). The vLLM path reproduces the endpoint behavior at 84 ms median per chunk, 5.4× faster than the transformers simulator used during development; latency claims made on $O(T^2)$ re-decode simulators overstate the cost of this approach by that factor.

Bounding every resident state matters: a 20 s force-flush fixes both the WER tail and the compute tail, and per-frame P95 stays flat through 8 concurrent sessions on a 0.15-GPU slice (a contention-caveated lower bound; the shape, not the constant, is the result). Re-measured on the released rank-32 checkpoint, architecturally identical to the endpoint model after merging, serving is unchanged: median per-chunk compute is 21 ms on an otherwise-idle GPU (the 84 ms above is under a co-tenant), and per-frame P95 stays flat through 16 concurrent sessions (77 ms at N=16, well inside the 500 ms budget).

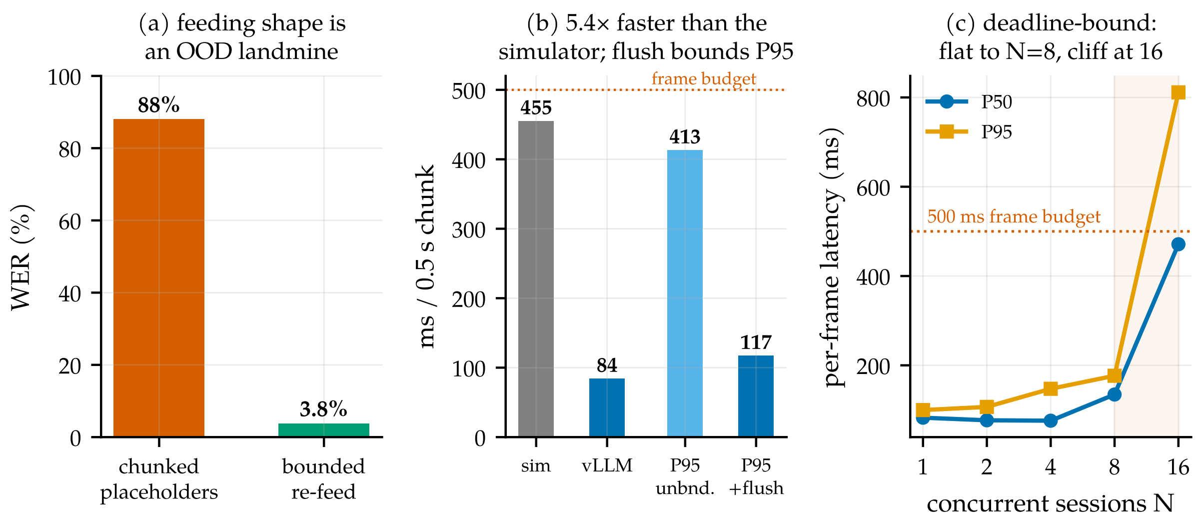

Figure 14. Serving on real infrastructure. (a) Feeding shape: per-chunk placeholders are out-of-distribution; bounded re-feed is correct. (b) Per-chunk compute on vLLM vs. the transformers simulator; the force-flush bounds the P95 tail. (c) Concurrent sessions per engine on a 0.15-GPU slice: flat to N=8, cliff at 16.